

CHAPTER 15

SEISMIC DESIGN REQUIREMENTS FOR NONBUILDING STRUCTURES

15.1 GENERAL

15.1.1 Nonbuilding Structures. Nonbuilding structures include all self-supporting structures that carry gravity loads and that may be required to resist the effects of earthquake, with the exception of building structures specifically excluded in Section 11.1.2 and other nonbuilding structures where specific seismic provisions have yet to be developed, and therefore, are not set forth in Chapter 15. Nonbuilding structures supported by the earth or supported by other structures shall be designed and detailed to resist the minimum lateral forces specified in this chapter. Design shall conform to the applicable requirements of other sections as modified by this section. Foundation design shall comply with the requirements of Sections 12.1.5, 12.13, and Chapter 14.

15.1.2 Design. The design of nonbuilding structures shall provide sufficient stiffness, strength, and ductility consistent with the requirements specified herein for buildings to resist the effects of seismic ground motions as represented by these design forces:

- Applicable strength and other design criteria shall be obtained from other portions of the seismic requirements of this standard or its reference documents.
- Where applicable strength and other design criteria are not contained in or referenced by the seismic requirements of this standard, such criteria shall be obtained from reference documents. Where reference documents define acceptance criteria in terms of allowable stresses as opposed to strength, the design seismic forces shall be obtained from this section and used in combination with other loads as specified in Section 2.4 of this standard and used directly with allowable stresses specified in the reference documents. Detailing shall be in accordance with the reference documents.

15.1.3 Structural Analysis Procedure Selection. Structural analysis procedures for nonbuilding structures that are similar to buildings shall be selected in accordance with Section 12.6. Nonbuilding structures that are not similar to buildings shall be designed using the equivalent lateral force procedure in accordance with Section 12.8, the linear dynamic analysis procedures in accordance with Section 12.9, the nonlinear response history analysis procedure in accordance with Chapter 16, or the procedure prescribed in the specific reference document.

EXCEPTIONS:

- For distributed mass cantilever structures listed in Table 15.4-2, including steel and reinforced concrete

stacks, chimneys, silos, skirt-supported vertical vessels, and steel tubular support structures for onshore wind turbine generator systems using the modal analysis procedure of Section 12.9.1 and the combined response for the modal base shear (V_t) at less than 85% of the calculated base shear (V) using the equivalent lateral force procedure, it is permitted to multiply the forces by $0.85V/V_t$ in lieu of the provisions of Section 12.9.1.

- For distributed mass cantilever structures listed in Table 15.4-2, including steel and reinforced concrete stacks, chimneys, silos, skirt-supported vertical vessels, and steel tubular support structures for onshore wind turbine generator systems using the modal analysis procedure of Section 12.9.1 and the combined response for the modal base shear (V_t) at less than $0.85C_sW$, and where C_s is determined in accordance with Eq. (12.8-6), drifts shall be multiplied by $0.85C_sW/V_t$ in lieu of the provisions of Section 12.9.

15.1.4 Nonbuilding Structures Sensitive to Vertical Ground Motions.

Tanks, vessels, hanging structures, and nonbuilding structures incorporating horizontal cantilevers shall use Section 11.9 to determine the vertical seismic design ground motion to be applied to the design of the nonbuilding structure. For these structures, the design forces in members and connections shall be determined by modal analysis performed in accordance with Section 12.9 or response history analysis performed in accordance with Chapter 12 or Chapter 16, except that the vertical ground motion component shall be included in the analysis using the spectra defined in Section 11.9. Alternatively, the equivalent static procedures of Section 12.8 and Section 15.4 are permitted to be used to determine the seismic design force with the vertical component motions. For tanks and vessels, vertical ground motions shall be applied as required by Section 15.7.2c. For hanging structures and nonbuilding structures incorporating horizontal cantilevers, the design vertical response spectral acceleration, S_{av} , shall be taken as the peak value from the response spectrum of Section 11.9. Alternatively, the design vertical response spectral acceleration, S_{av} , is permitted to be determined using the provisions of Section 11.9 using the vertical period of the structure determined by rational analysis. Horizontal seismic effects shall be combined with vertical seismic effects using the direction of loading criteria specified in Section 15.1.4.1 for hanging structures and structures incorporating horizontal cantilevers. The response modification factor, R , for use with vertical seismic design ground motions shall be taken as 1.0 except in the determination of hydrodynamic hoop forces in cylindrical tank walls. The determination of hydrodynamic hoop forces caused by vertical seismic design ground motions in

cylindrical tank walls shall comply with the requirements of Section 15.7.2c(2).

15.1.4.1 Direction of Loading Criteria for Nonbuilding Structures Sensitive to Vertical Ground Motions. The following orthogonal load combinations of horizontal and vertical seismic load effects shall be applied to hanging structures and structures incorporating horizontal cantilevers.

15.1.4.1.1 Strength. The directions of application of seismic forces used in the strength design of structure elements shall be those that produce the most critical load effects. This requirement is deemed satisfied if the structure elements are designed to the more stringent demands of the following load directions:

1. 100% of the forces for one horizontal direction plus 30% of the forces for the perpendicular horizontal direction plus 30% of the forces for the vertical direction. The combination requiring the maximum component strength shall be used.
2. 100% of the forces for the vertical direction plus 30% of the forces for a horizontal direction plus 30% of the forces for the perpendicular horizontal direction. The combination requiring the maximum component strength shall be used.

15.1.4.1.2 Overturning and Stability. The directions of application of seismic forces used in the evaluation of overturning stability and sliding of the structure shall be those that produce the most critical load effects. This requirement is deemed satisfied if the structures and their foundations are evaluated for overturning stability and sliding using the following load directions:

1. 100% of the forces for one horizontal direction plus 30% of the forces for the perpendicular horizontal direction plus 30% of the forces for the vertical direction. The combination requiring the maximum component strength shall be used.

15.2 THIS SECTION INTENTIONALLY LEFT BLANK; SEE SECTION 15.8

15.3 NONBUILDING STRUCTURES SUPPORTED BY OTHER STRUCTURES

Where nonbuilding structures identified in Table 15.4-2 are supported by other structures and nonbuilding structures are not part of the primary seismic force-resisting system, one of the following methods shall be used.

15.3.1 Less Than 25% Combined Weight Condition. For the condition where the weight of the nonbuilding structure is less than 25% of the combined effective seismic weights of the nonbuilding structure and supporting structure, the design seismic forces of the nonbuilding structure shall be determined in accordance with Chapter 13 where the values of R_p and a_p shall be determined in accordance with Section 13.1.6. The supporting structure shall be designed in accordance with the requirements of Chapter 12 or Section 15.5, as appropriate, with the weight of the nonbuilding structure considered in the determination of the effective seismic weight, W .

15.3.2 Greater Than or Equal to 25% Combined Weight Condition. For the condition where the weight of the nonbuilding structure is equal to or greater than 25% of the combined effective seismic weights of the nonbuilding structure and supporting structure, an analysis combining the structural

characteristics of both the nonbuilding structure and the supporting structures shall be performed to determine the seismic design forces as follows:

1. Where the fundamental period, T , of the nonbuilding structure is less than 0.06 s, the nonbuilding structure shall be considered a rigid element with appropriate distribution of its effective seismic weight. The supporting structure shall be designed in accordance with the requirements of Chapter 12 or Section 15.5 as appropriate, and the R value of the combined system is permitted to be taken as the R value of the supporting structural system. The nonbuilding structure and attachments shall be designed for the forces using the procedures of Chapter 13 where the value of R_p shall be taken as equal to the R value of the nonbuilding structure as set forth in Table 15.4-2, and a_p shall be taken as 1.0.
2. Where the fundamental period, T , of the nonbuilding structure is 0.06 s or greater, the nonbuilding structure and supporting structure shall be modeled together in a combined model with appropriate stiffness and effective seismic weight distributions. The combined structure shall be designed in accordance with Section 15.5 with the R value of the combined system taken as the lesser R value of the nonbuilding structure or the supporting structure. The nonbuilding structure and attachments shall be designed for the forces determined for the nonbuilding structure in the combined analysis.

15.3.3 Architectural, Mechanical, and Electrical Components. Architectural, mechanical, and electrical components supported by nonbuilding structures shall be designed in accordance with Chapter 13 of this standard.

15.4 STRUCTURAL DESIGN REQUIREMENTS

15.4.1 Design Basis. Nonbuilding structures that have specific seismic design criteria established in reference documents shall be designed using the standards as amended herein. Where reference documents are not cited herein, nonbuilding structures shall be designed in compliance with Sections 15.5 and 15.6 to resist minimum seismic lateral forces that are not less than the requirements of Section 12.8, with the following additions and exceptions:

1. The seismic force-resisting system shall be selected as follows:
 - a. For nonbuilding structures similar to buildings, a system shall be selected from among the types indicated in Table 12.2-1 or Table 15.4-1 subject to the system limitations and limits on structural height, h_n , based on the Seismic Design Category indicated in the table. The appropriate values of R , Ω_0 , and C_d indicated in the selected table shall be used in determining the base shear, element design forces, and design story drift as indicated in this standard. Design and detailing requirements shall comply with the sections referenced in the selected table.
 - b. For nonbuilding structures not similar to buildings, a system shall be selected from among the types indicated in Table 15.4-2 subject to the system limitations and limits on structural height, h_n , based on Seismic Design Category indicated in the table. The appropriate values of R , Ω_0 , and C_d indicated in Table 15.4-2 shall be used in determining the base shear, element design forces, and design story drift as indicated in this standard.

Table 15.4-1 Seismic Coefficients for Nonbuilding Structures Similar to Buildings

Nonbuilding Structure Type	Detailing Requirements	R	Ω_0	C_d	Structural System and Structural Height, h_n , Limits (ft) ^a				
					Seismic Design Category				
					B	C	D ^b	E ^b	F ^c
Steel storage racks	Sec. 15.5.3.1	4	2	3.5	NL	NL	NL	NL	NL
Steel cantilever storage racks hot-rolled steel									
Ordinary moment frame (cross-aisle)	15.5.3.2 and AISC 360	3	3	3	NL	NL	NP	NP	NP
Ordinary moment frame (cross-aisle) ^d	15.5.3.2 and AISC 341	2.5	2	2.5	NL	NL	NL	NL	NL
Ordinary braced frame (cross-aisle)	15.5.3.2 and AISC 360	3	3	3	NL	NL	NP	NP	NP
Ordinary braced frame (cross-aisle) ^d	15.5.3.2 and AISC 341	3.25	2	3.25	NL	NL	NL	NL	NL
Steel cantilever storage racks cold-formed steel ^e									
Ordinary moment frame (cross-aisle)	15.5.3.2 and AISI S100	3	3	3	NL	NL	NP	NP	NP
Ordinary moment frame (cross-aisle)	15.5.3.2 and AISI S100	1	1	1	NL	NL	NL	NL	NL
Ordinary braced frame (cross-aisle)	15.5.3.2 and AISI S100	3	3	3	NL	NL	NP	NP	NP
Building frame systems:									
Steel special concentrically braced frames	AISC 341	6	2	5	NL	NL	160	160	100
Steel ordinary concentrically braced frame	AISC 341	3 ^{3/4}	2	3 ^{3/4}	NL	NL	35 ^f	35 ^f	NP ^f
With permitted height increase	AISC 341	2 ^{1/2}	2	2 ^{1/2}	NL	NL	160	160	100
With unlimited height	AISC 360	1.5	1	1.5	NL	NL	NL	NL	NL
Moment-resisting frame systems:									
Steel special moment frames	AISC 341	8	3	5.5	NL	NL	NL	NL	NL
Special reinforced concrete moment frames ^g	ACI 318, including Chapter 18	8	3	5.5	NL	NL	NL	NL	NL
Steel intermediate moment frames:	AISC 341	4.5	3	4	NL	NL	35 ^{h,i}	NP ^{h,i}	NP ^{h,i}
With permitted height increase	AISC 341	2.5	2	2.5	NL	NL	160	160	100
With unlimited height	AISC 341	1.5	1	1.5	NL	NL	NL	NL	NL
Intermediate reinforced concrete moment frames:	ACI 318, including Chapter 18	5	3	4.5	NL	NL	NP	NP	NP
With permitted height increase	ACI 318, including Chapter 18	3	2	2.5	NL	NL	50	50	50
With unlimited height	ACI 318, including Chapter 18	0.8	1	1	NL	NL	NL	NL	NL
Steel ordinary moment frames:	AISC 341	3.5	3	3	NL	NL	NP ^{h,i}	NP ^{h,i}	NP ^{h,i}
With permitted height increase	AISC 341	2.5	2	2.5	NL	NL	100	100	NP ^{h,i}
With unlimited height	AISC 360	1	1	1	NL	NL	NL	NL	NL
Ordinary reinforced concrete moment frames:	ACI 318, excluding Chapter 18	3	3	2.5	NL	NP	NP	NP	NP
With permitted height increase	ACI 318, excluding Chapter 18	0.8	1	1	NL	NL	50	50	50

^aNL = no limit and NP = not permitted.

^bSee Section 12.2.5.4 for a description of seismic force-resisting systems limited to structures with a structural height, h_n , of 240 ft (73.2 m) or less.

^cSee Section 12.2.5.4 for seismic force-resisting systems limited to structures with a structural height, h_n , of 160 ft (48.8 m) or less.

^dThe column-to-base connection shall be designed to the lesser of M_n of the column or the factored moment at the base of the column for the seismic load case using the overstrength factor.

^eCold-formed sections that meet the requirements of AISC 341, Table D1.1, are permitted to be designed in accordance with AISC 341.

^fSteel ordinary braced frames are permitted in pipe racks up to 65 ft (20 m).

^gIn Section 2.3 of ACI 318, the definition of "special moment frame" includes precast and cast-in-place construction.

^hSteel ordinary moment frames and intermediate moment frames are permitted in pipe racks up to 65 ft (20 m) where the moment joints of field connections are constructed of bolted end plates.

ⁱSteel ordinary moment frames and intermediate moment frames are permitted in pipe racks up to 35 ft (11 m).

Design and detailing requirements shall comply with the sections referenced in Table 15.4-2.

- c. Where neither Table 15.4-1 nor Table 15.4-2 contains an appropriate entry, applicable strength and other design criteria shall be obtained from a reference document that is applicable to the specific type of nonbuilding structure. Design and detailing requirements shall comply with the reference document.

2. For nonbuilding systems that have an R value provided in Table 15.4-2, the minimum specified value in Eq. (12.8-5) shall be replaced by

$$C_s = 0.044S_{DS}I_e \quad (15.4-1)$$

The value of C_s shall not be taken as less than 0.03.

And for nonbuilding structures located where $S_1 \geq 0.6g$, the minimum specified value in Eq. (12.8-6) shall be replaced by

$$C_s = 0.8S_1/(R/I_e) \quad (15.4-2)$$

EXCEPTION: Tanks and vessels that are designed to AWWA D100; AWWA D103; API 650, Appendix E; and API 620, Appendix L, as modified by this standard, and stacks and chimneys that are designed to ACI 307 as modified by this standard, shall be subject to the larger of the minimum base shear value defined by the reference document or the value determined by replacing Eq. (12.8-5) with the following:

$$C_s = 0.044S_{DS}I_e \quad (15.4-3)$$

Table 15.4-2 Seismic Coefficients for Nonbuilding Structures Not Similar to Buildings

Nonbuilding Structure Type	Detailing Requirements ^c	R	Ω_0	C_d	Structural System and Structural Height, h_n , Limits (ft) ^{a,b}				
					Seismic Design Category				
					B	C	D	E	F
Elevated tanks, vessels, bins, or hoppers:									
On symmetrically braced legs (not similar to buildings)	Sec. 15.7.10	3	2 ^d	2.5	NL	NL	160	100	100
On unbraced legs or asymmetrically braced legs (not similar to buildings)	Sec. 15.7.10	2	2 ^d	2.5	NL	NL	100	60	60
Horizontal, saddle-supported welded steel vessels	Sec. 15.7.14	3	2 ^d	2.5	NL	NL	NL	NL	NL
Flat-bottom ground-supported tanks:	Sec. 15.7								
Steel or fiber-reinforced plastic:									
Mechanically anchored		3	2 ^d	2.5	NL	NL	NL	NL	NL
Self-anchored		2.5	2 ^d	2	NL	NL	NL	NL	NL
Reinforced or prestressed concrete:									
Reinforced nonsliding base		2	2 ^d	2	NL	NL	NL	NL	NL
Anchored flexible base		3.25	2 ^d	2	NL	NL	NL	NL	NL
Unanchored and unconstrained flexible base		1.5	1.5 ^d	1.5	NL	NL	NL	NL	NL
All other		1.5	1.5 ^d	1.5	NL	NL	NL	NL	NL
Cast-in-place concrete silos that have walls continuous to the foundation	Sec. 15.6.2	3	1.75	3	NL	NL	NL	NL	NL
All other reinforced masonry structures not similar to buildings detailed as intermediate reinforced masonry shear walls	Sec. 14.4.1 ^e	3	2	2.5	NL	NL	50	50	50
All other reinforced masonry structures not similar to buildings detailed as ordinary reinforced masonry shear walls	Sec. 14.4.1	2	2.5	1.75	NL	160	NP	NP	NP
All other nonreinforced masonry structures not similar to buildings	Sec. 14.4.1	1.25	2	1.5	NL	NP	NP	NP	NP
Concrete chimneys and stacks	Sec. 15.6.2 and ACI 307	2	1.5	2.0	NL	NL	NL	NL	NL
Steel chimneys and stacks	15.6.2 and ASME STS-1	2	2	2	NL	NL	NL	NL	NL
All steel and reinforced concrete distributed mass cantilever structures not otherwise covered herein, including stacks, chimneys, silos, skirt-supported vertical vessels; single-pedestal or skirt-supported	Sec. 15.6.2								
Welded steel	Sec. 15.7.10	2	2 ^d	2	NL	NL	NL	NL	NL
Welded steel with special detailing ^f	Secs. 15.7.10 and 15.7.10.5 a and b	3	2 ^d	2	NL	NL	NL	NL	NL
Prestressed or reinforced concrete	Sec. 15.7.10	2	2 ^d	2	NL	NL	NL	NL	NL
Prestressed or reinforced concrete with special detailing	Secs. 15.7.10 and ACI 318, Chapter 18, Secs. 18.2 and 18.10	3	2 ^d	2	NL	NL	NL	NL	NL
Trussed towers (freestanding or guyed), guyed stacks, and chimneys	Sec. 15.6.2	3	2	2.5	NL	NL	NL	NL	NL
Steel tubular support structures for onshore wind turbine generator systems	Sec. 15.6.7	1.5	1.5	1.5	NL	NL	NL	NL	NL
Cooling towers:									
Concrete or steel		3.5	1.75	3	NL	NL	NL	NL	NL
Wood frames		3.5	3	3	NL	NL	NL	50	50
Telecommunication towers:	Sec. 15.6.6								
Truss: Steel		3	1.5	3	NL	NL	NL	NL	NL
Pole: Steel		1.5	1.5	1.5	NL	NL	NL	NL	NL
Wood		1.5	1.5	1.5	NL	NL	NL	NL	NL
Concrete		1.5	1.5	1.5	NL	NL	NL	NL	NL
Frame: Steel		3	1.5	1.5	NL	NL	NL	NL	NL
Wood		1.5	1.5	1.5	NL	NL	NL	NL	NL
Concrete		2	1.5	1.5	NL	NL	NL	NL	NL
Amusement structures and monuments	Sec. 15.6.3	2	2	2	NL	NL	NL	NL	NL
Inverted pendulum type structures (except elevated tanks, vessels, bins, and hoppers)	Sec. 12.2.5.3	2	2	2	NL	NL	NL	NL	NL
Ground-supported cantilever walls or fences	Sec. 15.6.8	1.25	2	2.5	NL	NL	NL	NL	NL
Signs and billboards		3.0	1.75	3	NL	NL	NL	NL	NL
All other self-supporting structures, tanks, or vessels not covered above or by reference standards that are not similar to buildings		1.25	2	2.5	NL	NL	50	50	50

^aNL = no limit and NP = not permitted.

^bFor the purpose of height limit determination, the height of the structure shall be taken as the height to the top of the structural frame making up the primary seismic force-resisting system.

^cIf a section is not indicated in the detailing requirements column, no specific detailing requirements apply.

^dSee Section 15.7.3.a for the application of the overstrength factors, Ω_0 , for tanks and vessels.

^eDetailed with an essentially complete vertical load-carrying frame.

^fSections 15.7.10.5.a and 15.7.10.5.b shall be applied for any risk category.

The value of C_s shall not be taken as less than 0.01.
And for nonbuilding structures located where $S_1 \geq 0.6g$, the minimum specified value in Eq. (12.8-6) shall be replaced by

$$C_s = 0.5S_1/(R/I_e) \quad (15.4-4)$$

Minimum base shear requirements need not apply to the convective (sloshing) component of liquid in tanks.

3. The Importance Factor, I_e , shall be as set forth in Section 15.4.1.1.
4. The vertical distribution of the lateral seismic forces in nonbuilding structures covered by this section shall be determined:
 - a. Using the requirements of Section 12.8.3, or
 - b. Using the procedures of Section 12.9.1, or
 - c. In accordance with the reference document applicable to the specific nonbuilding structure.
5. Provided that the mass locations for the structure, any contents, and any supported structural or nonstructural elements (including but not limited to piping and stairs) that could contribute to the mass or stiffness of the structure are accounted for and quantified in the analysis, the accidental torsion requirements of Section 12.8.4.2 need not be accounted for:
 - a. Rigid nonbuilding structures, or
 - b. Nonbuilding structures not similar to buildings designed with R values less than or equal to 3.5, or
 - c. Nonbuilding structures similar to buildings with R values less than or equal to 3.5, provided that one of the following conditions is met:
 - i. The calculated center of rigidity at each diaphragm is greater than 5% of the plan dimension of the diaphragm in each direction from the calculated center of mass of the diaphragm, or
 - ii. The structure does not have a horizontal torsional irregularity type 1A or 1B and the structure has at least two lines of lateral resistance in each of two major axis directions. At least one line of lateral resistance shall be provided a distance of not less than 20% of the structure's plan dimension from the center of mass on each side of the center of mass.

In addition, structures designed to this section shall be analyzed using a 3D representation in accordance with Section 12.7.3.

6. For nonbuilding structural systems containing liquids, gases, and granular solids supported at the base as defined in Section 15.7.1, the minimum seismic design force shall not be less than that required by the reference document for the specific system.
7. Where a reference document provides a basis for the earthquake-resistant design of a particular type of nonbuilding structure covered by Chapter 15, such a standard shall not be used unless the following limitations are met:
 - a. The seismic ground accelerations and seismic coefficients shall be in conformance with the requirements of Section 11.4.
 - b. The values for total lateral force and total base overturning moment used in design shall not be less than 80% of the base shear value and overturning moment, each adjusted for the effects of soil–structure interaction that is obtained using this standard.
8. The base shear is permitted to be reduced in accordance with Section 19.2 to account for the effects of foundation

damping from soil–structure interaction. In no case shall the reduced base shear be less than $0.7V$.

9. Unless otherwise noted in Chapter 15, the effects on the nonbuilding structure caused by gravity loads and seismic forces shall be combined in accordance with the factored load combinations as presented in Section 2.3.
10. Where specifically required by Chapter 15, the design seismic force on nonbuilding structures shall be as defined in Section 12.4.3.

15.4.1.1 Importance Factor. The Importance Factor, I_e , and risk category for nonbuilding structures are based on the relative hazard of the contents and the function. The value of I_e shall be the largest value determined by the following:

- a. applicable reference document listed in Chapter 23,
- b. the largest value as selected from Table 1.5-2, or
- c. as specified elsewhere in Chapter 15.

15.4.2 Rigid Nonbuilding Structures. Nonbuilding structures that have a fundamental period, T , less than 0.06 s, including their anchorages, shall be designed for the lateral force obtained from the following:

$$V = 0.30S_{DS}WI_e \quad (15.4-5)$$

where

V = the total design lateral seismic base shear force applied to a nonbuilding structure;

S_{DS} = the site design response acceleration as determined from Section 11.4.5;

W = nonbuilding structure operating weight; and

I_e = the Importance Factor determined in accordance with Section 15.4.1.1.

The force shall be distributed with height in accordance with Section 12.8.3.

15.4.3 Loads. The seismic effective weight W for nonbuilding structures shall include the dead load and other loads as defined for structures in Section 12.7.2. For purposes of calculating design seismic forces in nonbuilding structures, W also shall include all normal operating contents for items such as tanks, vessels, bins, hoppers, and the contents of piping. W shall include snow and ice loads where these loads constitute 25% or more of W or where required by the Authority Having Jurisdiction based on local environmental characteristics.

15.4.4 Fundamental Period. The fundamental period of the nonbuilding structure shall be determined using the structural properties and deformation characteristics of the resisting elements in a properly substantiated analysis as indicated in Section 12.8.2. Alternatively, the fundamental period T is permitted to be computed from the following equation:

$$T = 2\pi \sqrt{\frac{\sum_{i=1}^n w_i \delta_i^2}{g \sum_{i=1}^n f_i \delta_i}} \quad (15.4-6)$$

The values of f_i represent any lateral force distribution in accordance with the principles of structural mechanics. The elastic deflections, δ_i , shall be calculated using the applied lateral forces, f_i . Eqs. (12.8-7), (12.8-8), (12.8-9), and (12.8-10) shall not be used for determining the period of a nonbuilding structure.

15.4.5 Drift Limit. The drift limit of Section 12.12.1 need not apply to nonbuilding structures if a rational analysis indicates that they can be exceeded without adversely affecting structural stability or attached or interconnected components and elements such as walkways and piping. P-delta effects shall be considered where they are critical to the function or stability of the structure. P-delta effects, when considered, shall be based on displacements determined by an elastic analysis multiplied by C_d/I_e using the appropriate C_d value from Tables 12.2-1, 15.4-1, or 15.4-2.

15.4.6 Materials Requirements. The requirements regarding specific materials in Chapter 14 shall be applicable unless specifically exempted in Chapter 15.

15.4.7 Drift, Deflection, and Structure Separation. Drift, deflection, and structure separation calculated using strength level seismic forces shall be determined in accordance with this standard unless specifically amended in Chapter 15.

15.4.8 Site-Specific Response Spectra. Where required by a reference document or the Authority Having Jurisdiction, specific types of nonbuilding structures shall be designed for site-specific criteria that account for local seismicity and geology, expected recurrence intervals, and magnitudes of events from known seismic hazards (see Section 11.4.8 of this standard). If a longer recurrence interval is defined in the reference document for the nonbuilding structure, such as liquefied natural gas (LNG) tanks (NFPA 59A), the recurrence interval required in the reference document shall be used.

15.4.9 Anchors in Concrete or Masonry

15.4.9.1 Anchors in Concrete. Anchors in concrete used for nonbuilding structure anchorage shall be designed in accordance with Chapter 17 of ACI 318.

15.4.9.2 Anchors in Masonry. Anchors in masonry used for nonbuilding structure anchorage shall be designed in accordance with TMS 402. Anchors shall be designed to be governed by the tensile or shear strength of a ductile steel element.

EXCEPTION: Anchors shall be permitted to be designed so that either:

- the attachment that the anchor is connecting to the structure undergoes ductile yielding at a load level corresponding to anchor forces not greater than the design strength of the anchor, or
- the anchors shall be designed to resist the load combinations in accordance with Section 12.4.3 including Ω_0 as given in Tables 15.4-1 and 15.4-2.

15.4.9.3 Post-installed Anchors in Concrete and Masonry. Post-installed anchors in concrete shall be prequalified for seismic applications in accordance with ACI 355.2 or other approved qualification procedures. Post-installed anchors in masonry shall be prequalified for seismic applications in accordance with approved qualification procedures.

15.4.9.4 ASTM F1554 Anchors. When ASTM F1554 Grade 36 anchors are specified and are designed as ductile anchors in accordance with ACI 318, Section 17.2.3.4.3(a), or where the design must meet the requirements of Section 15.7.5 or Section 15.7.11.7b, substitution of weldable ASTM F1554 Grade 55 (with Supplementary Requirement S1) anchors shall be prohibited.

15.4.10 Requirements for Nonbuilding Structure Foundations on Liquefiable Sites. Nonbuilding structure

foundations on liquefiable sites shall comply with Section 12.13.9 and the requirements of Section 15.4.10.1.

15.4.10.1 Nonbuilding Structures on Shallow Foundations. Nonbuilding structures shall not be permitted to be supported on shallow foundations at liquefiable sites unless it can be demonstrated that the structure's foundation, superstructure, and connecting systems can be designed to accommodate the soil strength loss, lateral spreading, and total and differential settlements induced by MCE_G earthquake ground motions indicated in the geotechnical investigation report.

15.5 NONBUILDING STRUCTURES SIMILAR TO BUILDINGS

15.5.1 General. Nonbuilding structures similar to buildings as defined in Section 11.2 shall be designed in accordance with this standard as modified by this section and the specific reference documents. This general category of nonbuilding structures shall be designed in accordance with the seismic requirements of this standard and the applicable portions of Section 15.4. The combination of load effects, E , shall be determined in accordance with Section 12.4.

15.5.2 Pipe Racks

15.5.2.1 Design Basis. In addition to the requirements of Section 15.5.1, pipe racks supported at the base of the structure shall be designed to meet the force requirements of Section 12.8 or 12.9.1. Displacements of the pipe rack and potential for interaction effects (pounding of the piping system) shall be considered using the amplified deflections obtained from the following equation:

$$\delta_x = \frac{C_d \delta_{xe}}{I_e} \quad (15.5-1)$$

where

C_d = deflection amplification factor in Table 15.4-1;

δ_{xe} = deflections determined using the prescribed seismic design forces of this standard; and

I_e = Importance Factor determined in accordance with Section 15.4.1.1.

See Section 13.6.2 for the design of piping systems and their attachments. Friction resulting from gravity loads shall not be considered to provide resistance to seismic forces.

15.5.3 Storage Racks. Storage racks constructed from steel supported at or below grade shall be designed in accordance with Sections 15.5.3.1 or 15.5.3.2, as applicable, and the requirements in Section 15.5.3.3.

15.5.3.1 Steel Storage Racks. Steel storage racks supported at or below grade shall be designed in accordance with ANSI/RMI MH 16.1 and its force and displacement requirements, except as follows.

15.5.3.1.1 Modify Section 7.1.2 of ANSI/RMI MH 16.1 as follows:

7.1.2 Base Plate Design

Once the required bearing area has been determined from the allowable bearing stress, F_p' , the minimum thickness of the base plate is determined by rational analysis or by appropriate test using a test load 1.5 times the ASD design load or the factored LRFD load. Design forces that include seismic loads for anchorage of steel storage racks to concrete or masonry shall be

determined using load combinations with overstrength provided in Section 12.4.3.1 of ASCE/SEI 7. The overstrength factor shall be taken as 2.0.

Anchorage of steel storage racks to concrete shall be in accordance with the requirements of Section 15.4.9 of ASCE/SEI 7. Upon request, information shall be given to the owner or the owner's agent on the location, size, and pressures under the column base plates of each type of upright frame in the installation. When rational analysis is used to determine base plate thickness, and other applicable standards do not apply, the base plate shall be permitted to be designed for the following loading conditions, where applicable: (balance of section unchanged)

15.5.3.1.2 Modify Section 7.1.4 of ANSI/RMI MH 16.1 as follows:

7.1.4 Shims

Shims may be used under the base plate to maintain the plumbness and/or levelness of the storage rack. The shims shall be made of a material that meets or exceeds the design bearing strength (LRFD) or allowable bearing strength (ASD) of the floor. The shim size and location under the base plate shall be equal to or greater than the required base plate size and location.

~~In no case shall the total thickness of a shim stack under a base plate exceed six times the diameter of the largest anchor bolt used in that base.~~

~~Shims stacks having a total thickness greater than two and less than or equal to six times the anchor bolt diameter under bases with only one anchor bolt shall be interlocked or welded together in a fashion that is capable of transferring all the shear forces at the base.~~

~~Shims stacks having a total thickness of less than or equal to two times the anchor bolt diameter need not be interlocked or welded together.~~

Bending in the anchor associated with shims or grout under the base plate shall be taken into account in the design of anchor bolts.

15.5.3.2 Steel Cantilevered Storage Racks. Steel cantilevered storage racks supported at or below grade shall be designed in accordance with ANSI/RMI MH 16.3, its force and displacement requirements, and the seismic design ground motion values determined according to Section 11.4, except as follows:

15.5.3.2.1 Modify Section 8.5.1 of ANSI/RMI MH 16.3 as follows:

8.5.1 Anchor Bolt Design

Anchorage of steel cantilevered storage racks to concrete shall be in accordance with the requirements of Section 15.4.9 of ASCE/SEI 7. The redundancy factor in the load combinations in Section 2.1 and 2.2 shall be 1.0. Design forces that include seismic loads for anchorage of steel cantilevered storage racks to concrete or masonry shall be determined using load combinations with overstrength provided in Sections 2.3.6 or 2.4.5 of ASCE/SEI 7.

If shims are used under the base plate to maintain the plumbness and/or levelness of the steel cantilevered storage rack, the shims stacks shall be interlocked or welded together in a fashion that is capable of transferring all the shear forces at the base. Bending in the anchor associated with shims or grout under the base plate shall be taken into account in the design of anchor bolts.

15.5.3.3 Alternative. As an alternative to ANSI/RMI MH 16.1 or 16.3, as modified above, storage racks shall be permitted to be

designed in accordance with the requirements of Sections 15.1, 15.2, 15.3, 15.5.1, and 15.5.3.3.1 through 15.5.3.3.4 of this standard.

15.5.3.3.1 General Requirements. Storage racks shall satisfy the force requirements of this section. The Importance Factor, I_e , for storage racks in structures open to the public, such as warehouse retail stores, shall be taken equal to 1.5.

EXCEPTION: Steel storage racks supported at the base are permitted to be designed as structures with an R of 4, provided that the seismic requirements of this standard are met. Higher values of R are permitted to be used where the detailing requirements of reference documents listed in Section 14.1.1 are met. The Importance Factor, I_e , for steel storage racks in structures open to the public, such as warehouse retail stores, shall be taken as equal to 1.5.

15.5.3.3.2 Operating Weight. Storage racks shall be designed for each of the following conditions of operating weight, W or W_p .

- a. Weight of the rack plus every storage level loaded to 67% of its rated load capacity.
- b. Weight of the rack plus the highest storage level only loaded to 100% of its rated load capacity.

The design shall consider the actual height of the center of mass of each storage load component.

15.5.3.3.3 Vertical Distribution of Seismic Forces. For all storage racks, the vertical distribution of seismic forces shall be as specified in Section 12.8.3 and in accordance with the following:

- a. The base shear, V , of the typical structure shall be the base shear of the steel storage rack where loaded in accordance with Section 15.5.3.2.
- b. The base of the structure shall be the floor supporting the steel storage rack. Each steel storage level of the rack shall be treated as a level of the structure with heights h_i and h_x measured from the base of the structure.
- c. The factor k is permitted to be taken as 1.0.

15.5.3.3.4 Seismic Displacements. Storage rack installations shall accommodate the seismic displacement of the steel storage racks and their contents relative to all adjacent or attached components and elements. The assumed total relative displacement for steel storage racks shall be not less than 5% of the structural height above the base, h_n , unless a smaller value is justified by test data or analysis in accordance with Section 11.1.4.

15.5.4 Electrical Power-Generating Facilities

15.5.4.1 General. Electrical power-generating facilities are power plants that generate electricity by steam turbines, combustion turbines, diesel generators, or similar turbo machinery.

15.5.4.2 Design Basis. In addition to the requirements of Section 15.5.1, electrical power-generating facilities shall be designed using this standard and the appropriate factors contained in Section 15.4.

15.5.5 Structural Towers for Tanks and Vessels

15.5.5.1 General. In addition to the requirements of Section 15.5.1, structural towers that support tanks and vessels and are not integral with the tank shall be designed to meet the requirements of Section 15.3. In addition, the following special considerations shall be included:

- a. The distribution of the lateral base shear from the tank or vessel onto the supporting structure shall consider the relative stiffness of the tank and resisting structural elements.
- b. The distribution of the vertical reactions from the tank or vessel onto the supporting structure shall consider the relative stiffness of the tank and resisting structural elements. Where the tank or vessel is supported on grillage beams, the calculated vertical reaction caused by weight and overturning shall be increased at least 20% to account for nonuniform support. The grillage beam and vessel attachment shall be designed for this increased design value.
- c. Seismic displacements of the tank and vessel shall consider the deformation of the support structure where determining P-delta effects or evaluating required clearances to prevent pounding of the tank on the structure. P-delta effects shall be based on displacements determined by an elastic analysis multiplied by C_d/I_e using the appropriate C_d value from Table 15.4-2.

Tanks and vessels supported by structural towers that are integral to the tank or vessel shall be designed according to Section 15.7.10.1.

15.5.6 Piers and Wharves

15.5.6.1 General. Piers and wharves are structures located in waterfront areas that project into a body of water or that parallel the shoreline.

15.5.6.2 Design Basis. In addition to the requirements of Section 15.5.1, piers and wharves that are accessible to the general public, such as cruise ship terminals and piers with retail or commercial offices or restaurants, shall be designed to comply with this standard. Piers and wharves that are not accessible to the general public are beyond the scope of this section.

The design shall account for the effects of liquefaction and soil failure collapse mechanisms and shall consider all applicable marine loading combinations, such as mooring, berthing, wave, and current on piers and wharves as required. Structural detailing shall consider the effects of the marine environment.

15.6 GENERAL REQUIREMENTS FOR NONBUILDING STRUCTURES NOT SIMILAR TO BUILDINGS

Nonbuilding structures that do not have lateral and vertical seismic force-resisting systems that are similar to buildings shall be designed in accordance with this standard as modified by this section and the specific reference documents. Loads and load distributions shall not be less demanding than those determined in this standard. The combination of earthquake load effects, E , shall be determined in accordance with Section 12.4.2.

EXCEPTION: The redundancy factor, ρ , per Section 12.3.4 shall be taken as 1.

15.6.1 Earth-Retaining Structures. This section applies to all earth-retaining structures assigned to Seismic Design Category D, E, or F. The lateral earth pressures caused by earthquake ground motions shall be determined in accordance with Section 11.8.3. The risk category shall be determined by the proximity of the earth-retaining structure to other buildings and structures. If failure of the earth-retaining structure would affect the adjacent building or structure, the risk category shall not be less than that of the adjacent building or structure.

Earth-retaining walls are permitted to be designed for seismic loads as either yielding or nonyielding walls. Cantilevered

reinforced concrete or masonry retaining walls shall be assumed to be yielding walls and shall be designed as simple flexural wall elements.

15.6.2 Chimneys and Stacks.

15.6.2.1 General. Chimneys and stacks are permitted to be either lined or unlined and shall be constructed from concrete, steel, or masonry. Steel stacks, concrete stacks, steel chimneys, concrete chimneys, and liners shall be designed to resist seismic lateral forces determined from a substantiated analysis using reference documents. Interaction of the stack or chimney with the liners shall be considered. A minimum separation shall be provided between the liner and chimney equal to C_d times the calculated differential lateral drift.

15.6.2.2 Concrete Chimneys and Stacks. Concrete chimneys and stacks shall be designed in accordance with the requirements of ACI 307 except that (1) the design base shear shall be determined based on Section 15.4.1 of this standard, (2) the seismic coefficients shall be based on the values provided in Table 15.4-2, and (3) openings shall be detailed as required below. When modal response spectrum analysis is used for design, the procedures of Section 12.9 shall be permitted to be used.

For concrete chimneys and stacks assigned to Seismic Design Category D, E, and F, splices for vertical rebar shall be staggered such that no more than 50% of the bars are spliced at any section and alternate lap splices are staggered by the development length. In addition, where the loss of cross-sectional area is greater than 10%, cross sections in the regions of breaches/openings shall be designed and detailed for vertical force, shear force, and bending moment demands along the vertical direction, determined for the affected cross section using an overstrength factor of 1.5. The region where the overstrength factor applies shall extend above and below the opening(s) by a distance equal to half of the width of the largest opening in the affected region. Appropriate reinforcement development lengths shall be provided beyond the required region of overstrength. The jamb regions around each opening shall be detailed using the column tie requirements in Section 10.7.6 of ACI 318. Such detailing shall extend for a jamb width of a minimum of two times the wall thickness and for a height of the opening height plus twice the wall thickness above and below the opening but no less than the development length of the longitudinal bars. Where the existence of a footing or base mat precludes the ability to achieve the extension distance below the opening and within the stack, the jamb reinforcing shall be extended and developed into the footing or base mat. The percentage of longitudinal reinforcement in jamb regions shall meet the requirements of Section 10.6.1.1 of ACI 318 for compression members.

15.6.2.3 Steel Chimneys and Stacks. Steel chimneys and stacks shall be designed in accordance with the requirements of ASME STS-1 except that (1) the design base shear shall be determined based on Section 15.4.1 of this standard, (2) the seismic coefficients shall be based on the values provided in Table 15.4-2, and (3) ASME STS-1, Section 4.3.5, and Nonmandatory Appendix D Table D-1, Table D-2, Fig. D-2, and Fig. D-2a shall not be used. When modal response spectrum analysis is used for design, the procedures of Section 12.9 shall be permitted to be used.

15.6.3 Amusement Structures. Amusement structures are permanently fixed structures constructed primarily for the conveyance and entertainment of people. Amusement structures shall be designed to resist seismic lateral forces determined in accordance with Section 15.4.

15.6.4 Special Hydraulic Structures. Special hydraulic structures are structures that are contained inside liquid-containing structures. These structures are exposed to liquids on both wall surfaces at the same head elevation under normal operating conditions. Special hydraulic structures are subjected to out-of-plane forces only during an earthquake where the structure is subjected to differential hydrodynamic fluid forces. Examples of special hydraulic structures include separation walls, baffle walls, weirs, and other similar structures.

15.6.4.1 Design Basis. Special hydraulic structures shall be designed for out-of-phase movement of the fluid. Unbalanced forces from the motion of the liquid must be applied simultaneously “in front of” and “behind” these elements.

Structures subject to hydrodynamic pressures induced by earthquakes shall be designed for rigid body and sloshing liquid forces and their own inertia force. The height of sloshing shall be determined and compared with the freeboard height of the structure. Interior elements, such as baffles or roof supports, also shall be designed for the effects of unbalanced forces and sloshing.

15.6.5 Secondary Containment Systems. Secondary containment systems, such as impoundment dikes and walls, shall meet the requirements of the applicable standards for tanks and vessels and the Authority Having Jurisdiction.

Secondary containment systems shall be designed to withstand the effects of the maximum considered earthquake ground motion where empty and two-thirds of the maximum considered earthquake ground motion where full including all hydrodynamic forces as determined in accordance with the procedures of Section 11.4. Where determined by the risk assessment required by Section 1.5.3 or by the Authority Having Jurisdiction that the site may be subject to aftershocks of the same magnitude as the maximum considered motion, secondary containment systems shall be designed to withstand the effects of the maximum considered earthquake ground motion where full including all hydrodynamic forces as determined in accordance with the procedures of Section 11.4.

15.6.5.1 Freeboard. Sloshing of the liquid within the secondary containment area shall be considered in determining the height of the impoundment. Where the primary containment has not been designed with a reduction in the structure category (i.e., no reduction in Importance Factor I_e) as permitted by Section 1.5.3, no freeboard provision is required. Where the primary containment has been designed for a reduced structure category (i.e., Importance Factor I_e reduced) as permitted by Section 1.5.3, a minimum freeboard, δ_s , shall be provided where

$$\delta_s = 0.42DS_{ac} \quad (15.6-1)$$

where S_{ac} is the spectral acceleration of the convective component and is determined according to the procedures of Section 15.7.6.1 using 0.5% damping. For circular impoundment dikes, D shall be taken as the diameter of the impoundment dike. For rectangular impoundment dikes, D shall be taken as the plan dimension of the impoundment dike, L , for the direction under consideration.

15.6.6 Telecommunication Towers. Self-supporting and guyed telecommunication towers shall be designed to resist seismic lateral forces determined in accordance with Section 15.4.

15.6.7 Steel Tubular Support Structures for Onshore Wind Turbine Generator Systems. Steel tubular support structures for onshore wind turbine generator systems shall be designed to

resist seismic lateral forces determined in accordance with Section 15.4.

15.6.8 Ground-Supported Cantilever Walls or Fences

15.6.8.1 General. Ground-supported cantilever walls or fences 6 ft (1.83 m) or greater in height shall satisfy the requirements of this section. Earth-retaining structures shall comply with Section 15.6.1.

15.6.8.2 Design Basis. Walls or fences shall be designed to resist earthquake ground motions in accordance with Section 15.4. Detailed plain and ordinary plain concrete or masonry walls or fences and ordinary plain autoclaved aerated concrete (AAC) walls or fences are not permitted in Seismic Design Category C, D, E, and F.

15.7 TANKS AND VESSELS

15.7.1 General. This section applies to all tanks, vessels, bins, silos, and similar containers storing liquids, gases, and granular solids supported at the base (hereafter referred to generically as “tanks and vessels”). Tanks and vessels covered herein include reinforced concrete, prestressed concrete, steel, aluminum, and fiber-reinforced plastic materials. Tanks supported on elevated levels in buildings shall be designed in accordance with Section 15.3.

15.7.2 Design Basis. Tanks and vessels storing liquids, gases, and granular solids shall be designed in accordance with this standard and shall be designed to meet the requirements of the applicable reference documents listed in Chapter 23. Resistance to seismic forces shall be determined from a substantiated analysis based on the applicable reference documents listed in Chapter 23.

- Damping for the convective (sloshing) force component shall be taken as 0.5%.
- Impulsive and convective components shall be combined by the direct sum or the square root of the sum of the squares (SRSS) method where the modal periods are separated. If significant modal coupling may occur, the complete quadratic combination (CQC) method shall be used.
- Vertical earthquake forces shall be considered in accordance with the applicable reference document. If the reference document permits the user the option of including or excluding the vertical earthquake force to comply with this standard, it shall be included. For tanks and vessels not covered by a reference document, the forces caused by the vertical acceleration shall be defined as follows:
 - Hydrodynamic vertical and lateral forces in non-cylindrical tank walls: The increase in hydrostatic pressures caused by the vertical excitation of the contained liquid shall correspond to an effective increase in unit weight, γ_L , of the stored liquid equal to $0.4S_{av}\gamma_L$, where S_{av} is taken as the peak of the vertical response spectrum defined in Section 11.9.
 - Hydrodynamic hoop forces in cylindrical tank walls: In a cylindrical tank wall, the hoop force per unit height, N_h , at height y from the base, associated with the vertical excitation of the contained liquid, shall be computed in accordance with Eq. (15.7-1). The hoop force associated with the vertical excitation of the liquid shall be combined with the impulsive and convective components by the direct sum or SRSS method:

$$N_h = \frac{S_{av}}{R} \gamma_L (H_L - y) \left(\frac{D_t}{2} \right) \quad (15.7-1)$$

where

D_i = inside tank diameter;
 H_L = liquid height inside the tank;
 y = distance from base of the tank to height being investigated;
 γ_L = unit weight of stored liquid; and
 S_{av} = vertical seismic parameter from Section 11.9, determined natural period of vibration of vertical liquid motion.

- (3) Vertical inertia forces in cylindrical and rectangular tank walls: Vertical inertia forces associated with the vertical acceleration of the structure itself shall be taken as equal to $0.4S_{av}W$ where S_{av} is taken as the peak of the vertical response spectrum defined in Section 11.9.

15.7.3 Strength and Ductility. Structural members that are part of the seismic force-resisting system shall be designed to provide the following:

- a. Connections to seismic force-resisting elements, excluding anchors (bolts or rods) embedded in concrete, shall be designed to develop Ω_0 times the calculated connection design force. For anchors (bolts or rods) embedded in concrete, the design of the anchor embedment shall meet the requirements of Section 15.7.5. Additionally, the connection of the anchors to the tank or vessel shall be designed to develop the lesser of the strength of the anchor in tension as determined by the reference document or Ω_0 times the calculated anchor design force. The over-strength requirements of Section 12.4.3 and the Ω_0 values tabulated in Table 15.4-2 do not apply to the design of walls, including interior walls, of tanks or vessels.
- b. Penetrations, manholes, and openings in shell elements shall be designed to maintain the strength and stability of the shell to carry tensile and compressive membrane shell forces.
- c. Support towers for tanks and vessels, where the support tower is integral with the tank or vessel, with irregular bracing, unbraced panels, asymmetric bracing, or concentrated masses shall be designed using the requirements of

Section 12.3.2 for irregular structures. Support towers using chevron or eccentrically braced framing shall comply with the seismic requirements of this standard. Support towers using tension-only bracing shall be designed such that the full cross section of the tension element can yield during overload conditions.

- d. In support towers for tanks and vessels, where the support tower is integral with the tank or vessel, compression struts that resist the reaction forces from tension braces shall be designed to resist the lesser of the yield load of the brace, $A_g F_y$, or Ω_0 times the calculated tension load in the brace.
- e. The vessel stiffness relative to the support system (foundation, support tower, skirt, etc.) shall be considered in determining forces in the vessel, the resisting elements, and the connections.
- f. For concrete liquid-containing structures, system ductility and energy dissipation under unfactored loads shall not be allowed to be achieved by inelastic deformations to such a degree as to jeopardize the serviceability of the structure. Stiffness degradation and energy dissipation shall be allowed to be obtained either through limited microcracking, or by means of lateral force resistance mechanisms that dissipate energy without damaging the structure.

15.7.4 Flexibility of Piping Attachments. Design of piping systems connected to tanks and vessels shall consider the potential movement of the connection points during earthquakes and provide sufficient flexibility to avoid release of the product by failure of the piping system. The piping system and supports shall be designed so as not to impart significant mechanical loading on the attachment to the tank or vessel shell. Mechanical devices that add flexibility, such as bellows, expansion joints, and other flexible apparatus, are permitted to be used where they are designed for seismic displacements and defined operating pressure.

Unless otherwise calculated, the minimum displacements in Table 15.7-1 shall be assumed. For attachment points located above the support or foundation elevation, the displacements in Table 15.7-1 shall be increased to account for drift of the tank or vessel relative to the base of support. The piping system and tank connection shall also be designed to tolerate C_d times the displacements given in Table 15.7-1 without rupture, although

Table 15.7-1 Minimum Design Displacements for Piping Attachments

Condition	Displacements (in.) (mm)
Mechanically Anchored Tanks and Vessels	
Upward vertical displacement relative to support or foundation	1 (25.4)
Downward vertical displacement relative to support or foundation	0.5 (12.7)
Range of horizontal displacement (radial and tangential) relative to support or foundation	0.5 (12.7)
Self-Anchored Tanks or Vessels (at grade)	
Upward vertical displacement relative to support or foundation:	
If designed in accordance with a reference document as modified by this standard:	
Anchorage ratio less than or equal to 0.785 (indicates no uplift)	1 (25.4)
Anchorage ratio greater than 0.785 (indicates uplift)	4 (101.1)
If designed for seismic loads in accordance with this standard but not covered by a reference document:	
For tanks and vessels with a diameter less than 40 ft (12.2 m)	8 (202.2)
For tanks and vessels with a diameter equal to or greater than 40 ft (12.2 m)	12 (0.305)
Downward vertical displacement relative to support or foundation:	
For tanks with a ringwall/mat foundation	0.5 (12.7)
For tanks with a berm foundation	1 (25.4)
Range of horizontal displacement (radial and tangential) relative to support or foundation	2 (50.8)

Table 15.7-2 Anchorage Ratio

<i>J</i> Anchorage Ratio	Criteria
$J < 0.785$	No uplift under the design seismic overturning moment. The tank is self-anchored.
$0.785 < J < 1.54$	Tank is uplifting, but the tank is stable for the design load, provided that the shell compression requirements are satisfied. The tank is self-anchored.
$J > 1.54$	Tank is not stable and shall be mechanically anchored for the design load.

permanent deformations and inelastic behavior in the piping supports and tank shell are permitted. For attachment points located above the support or foundation elevation, the displacements in Table 15.7-1 shall be increased to account for drift of the tank or vessel. The values given in Table 15.7-1 do not include the influence of relative movements of the foundation and piping anchorage points caused by foundation movements (e.g., settlement or seismic displacements). The effects of the foundation movements shall be included in the piping system design, including the determination of the mechanical loading on the tank or vessel, and the total displacement capacity of the mechanical devices intended to add flexibility.

The anchorage ratio, *J*, for self-anchored tanks shall comply with the criteria shown in Table 15.7-2 and is defined as

$$J = \frac{M_{rw}}{D^2(w_t + w_a)} \quad (15.7-2)$$

where

$$w_t = \frac{W_s}{\pi D} + w_r \quad (15.7-3)$$

w_r = roof load acting on the shell [lb/ft (N/m)] of shell circumference. Only permanent roof loads shall be included. Roof live load shall not be included.

w_a = maximum weight of the tank contents that may be used to resist the shell overturning moment [lb/ft (N/m)] of shell circumference; w_a usually consists of an annulus of liquid limited by the bending strength of the tank bottom or annular plate.

M_{rw} = the overturning moment applied at the bottom of the shell caused by the seismic design loads [ft-lb (N-m)] (also known as the “ringwall moment”).

D = tank diameter [ft (m)].

W_s = total weight of tank shell [lb (N)].

15.7.5 Anchorage. Tanks and vessels at grade are permitted to be designed without anchorage where they meet the requirements for self-anchored tanks in reference documents. Tanks and vessels supported above grade on structural towers or building structures shall be anchored to the supporting structure.

The following special detailing requirements shall apply to steel tank and vessel anchor bolts in Seismic Design Category C, D, E, and F. Anchorage shall be in accordance with Section 15.4.9, whereby the anchor embedment into the concrete shall be designed to develop the steel strength of the anchor in tension. The steel strength of the anchor in tension shall be determined in accordance with ACI 318, Eq. 17.4.1.2. The anchor shall have a minimum gauge length of eight diameters. Post-installed anchors are permitted to be used in accordance

with Section 15.4.9.3 provided that the anchor embedment into the concrete is designed to develop the steel strength of the anchor in tension. In either case, the load combinations including overstrength of Section 12.4.3 are not to be used to size the anchor bolts for tanks and horizontal and vertical vessels.

15.7.6 Ground-Supported Storage Tanks for Liquids

15.7.6.1 General. Ground-supported, flat-bottom tanks storing liquids shall be designed to resist the seismic forces calculated using one of the following procedures:

- The base shear and overturning moment are calculated as if the tank and the entire contents are a rigid mass system per Section 15.4.2 of this standard.
- Tanks or vessels storing liquids in Risk Category IV, or with a diameter greater than 20 ft (6.1 m), shall be designed to consider the hydrodynamic pressures of the liquid in determining the equivalent lateral forces and lateral force distribution per the applicable reference documents listed in Chapter 23 and the requirements of Section 15.7.
- The force and displacement requirements of Section 15.4 are used.

The design of tanks storing liquids shall consider the impulsive and convective (sloshing) effects and their consequences on the tank, foundation, and attached elements. The impulsive component corresponds to the high-frequency amplified response to the lateral ground motion of the tank roof, the shell, and the portion of the contents that moves in unison with the shell. The convective component corresponds to the low-frequency amplified response of the contents in the fundamental sloshing mode. Damping for the convective component shall be 0.5% for the sloshing liquid unless otherwise defined by the reference document. The following definitions shall apply:

D_i = inside diameter of tank or vessel;

H_L = design liquid height inside the tank or vessel;

L = inside length of a rectangular tank, parallel to the direction of the earthquake force being investigated;

N_h = hydrodynamic hoop force per unit height in the wall of a cylindrical tank or vessel;

T_c = natural period of the first (convective) mode of sloshing;

T_i = fundamental period of the tank structure and impulsive component of the content;

V_i = base shear caused by impulsive component from weight of tank and contents;

V_c = base shear caused by the convective component of the effective sloshing mass;

y = distance from base of the tank to level being investigated; and

γ_L = unit weight of stored liquid.

The seismic base shear is the combination of the impulsive and convective components:

$$V = V_i + V_c \quad (15.7-4)$$

where

$$V_i = \frac{S_{ai} W_i}{(I_e)} \quad (15.7-5)$$

$$V_c = \frac{S_{ac} I_e}{1.5} W_c \quad (15.7-6)$$

where

S_{ai} = the spectral acceleration as a multiplier of gravity including the site impulsive components at period T_i and 5% damping.
 W_i = impulsive weight (impulsive component of liquid, roof and equipment, shell, bottom, and internal elements); and
 W_c = the portion of the liquid weight sloshing.

For $T_i \leq T_s$,

$$S_{ai} = S_{DS} \quad (15.7-7)$$

For $T_s < T_i \leq T_L$,

$$S_{ai} = \frac{S_{D1}}{T_i} \quad (15.7-8)$$

For $T_i > T_L$,

$$S_{ai} = \frac{S_{D1} T_L}{T_i^2} \quad (15.7-9)$$

NOTES:

- Where a reference document is used in which the spectral acceleration for the tank shell and the impulsive component of the liquid are independent of T_i , then $S_{ai} = S_{DS}$.
- Eqs. (15.7-8) and (15.7-9) shall not be less than the minimum values required in Section 15.4.1, Item 2, multiplied by R/I_e .
- Impulsive and convective seismic forces for tanks are permitted to be combined using the square root of the sum of the squares (SRSS) method in lieu of the direct sum method shown in Section 15.7.6 and its related subsections.

S_{ac} = the spectral acceleration of the sloshing liquid (convective component) based on the sloshing period T_c and 0.5% damping.

For $T_c \leq T_L$:

$$S_{ac} = \frac{1.5 S_{D1}}{T_c} \leq S_{DS} \quad (15.7-10)$$

For $T_c > T_L$:

$$S_{ac} = \frac{1.5 S_{D1} T_L}{T_c^2} \quad (15.7-11)$$

EXCEPTION: For $T_c > 4$ s, S_{ac} is permitted to be determined by a site-specific study using one or more of the following methods: (1) the procedures found in Chapter 21, provided that such procedures, which rely on ground-motion attenuation equations for computing response spectra, cover the natural period band containing T_c ; (2) ground-motion simulation methods that use seismological models of fault rupture and wave propagation; and (3) analysis of representative strong-motion accelerogram data with reliable long-period content extending to periods greater than T_c . Site-specific values of S_{ac} shall be based on one standard deviation determinations. However, in no case shall the value of S_{ac} be taken as less than the value determined in accordance with Eq. (15.7-11) using 50% of the mapped value of T_L from Chapter 22.

The 80% limit on S_a required by Sections 21.3 and 21.4 shall not apply to the determination of site-specific values of S_{ac} , which satisfy the requirements of this exception. In determining the value of S_{ac} , the value of T_L shall not be less than 4 s where

$$T_c = 2\pi \sqrt{\frac{D}{3.68g \tanh\left(\frac{3.68H}{D}\right)}} \quad (15.7-12)$$

and where

D = the tank diameter [ft (m)]

H = liquid height [ft (m)]

g = acceleration caused by gravity in consistent units;

15.7.6.1.1 Distribution of Hydrodynamic and Inertia Forces. Unless otherwise required by the appropriate reference document listed in Chapter 23, the method given in ACI 350.3 is permitted to be used to determine the vertical and horizontal distribution of the hydrodynamic and inertial forces on the walls of circular and rectangular tanks.

15.7.6.1.2 Sloshing. Sloshing of the stored liquid shall be taken into account in the seismic design of tanks and vessels in accordance with the following requirements:

- The height of the sloshing liquid, δ_s , above the product design height shall be computed using Eq. (15.7-13):

$$\delta_s = 0.42 D_i I_e S_{ac} \quad (15.7-13)$$

For cylindrical tanks, D_i shall be the inside diameter of the tank; for rectangular tanks, the term D_i shall be replaced by the longitudinal plan dimension of the tank, L , for the direction under consideration.

- For tanks in Risk Category IV, the value of the Importance Factor, I_e , used for freeboard determination only, shall be taken as 1.0.
- For tanks in Risk Categories I, II, and III, the value of T_L used for freeboard determination is permitted to be set equal to 4 s. The value of the Importance Factor, I_e , used for freeboard determination for tanks in Risk Categories I, II, and III shall be the value determined from Table 1.5-1.
- The effects of sloshing shall be accommodated by means of one of the following:
 - A minimum freeboard in accordance with Table 15.7-3.
 - A roof and supporting structure designed to contain the sloshing liquid in accordance with subsection e below.
 - Secondary containment is provided to control the product spill.
 - For open-top tanks or vessels only, an overflow spillway around the tank or vessel perimeter.

EXCEPTION: No minimum freeboard is required for open-top tanks where the following conditions are met:

- Contained fluid is not toxic, explosive, or highly toxic and has been approved by the Authority Having Jurisdiction as acceptable for product spill.

Table 15.7-3 Minimum Required Freeboard

Value of S_{DS}	Risk Category		
	I or II	III	IV
$S_{DS} < 0.33g$	Not required	Not required	δ_s
$S_{DS} \geq 0.33g$	Not required	$0.7\delta_s$	δ_s

2. Site-specific product spill prevention, control, and countermeasure plan (SPCC) has been developed and approved by the Authority Having Jurisdiction to properly handle resulting spill. The SPCC shall account for proper site drainage, infiltration, foundation scour, and protection of adjacent facilities from sloshing spill.
- e. If the sloshing is restricted because the freeboard is less than the computed sloshing height, then the roof and supporting structure shall be designed for an equivalent hydrostatic head equal to the computed sloshing height less the freeboard. In addition, the design of the tank shall use the confined portion of the convective (sloshing) mass as an additional impulsive mass.

15.7.6.1.3 Equipment and Attached Piping. Equipment, piping, and walkways or other appurtenances attached to the structure shall be designed to accommodate the displacements imposed by seismic forces. For piping attachments, see Section 15.7.4.

15.7.6.1.4 Internal Elements. The attachments of internal equipment and accessories that are attached to the primary liquid or pressure retaining shell or bottom or that provide structural support for major elements (e.g., a column supporting the roof rafters) shall be designed for the lateral loads caused by the sloshing liquid in addition to the inertial forces by a substantiated analysis method.

15.7.6.1.5 Sliding Resistance. The transfer of the total lateral shear force between the tank or vessel and the subgrade shall be considered:

- a. For flat-bottom steel tanks, the overall horizontal seismic shear force is permitted to be resisted by friction between the tank bottom and the foundation or subgrade. Storage tanks shall be designed such that sliding does not occur where the tank is full of stored product. The maximum calculated seismic base shear, V , shall not exceed

$$V < W \tan 30^\circ \quad (15.7-14)$$

W shall be determined using the effective seismic weight of the tank, roof, and contents after reduction for coincident vertical earthquake. Lower values of the friction factor shall be used if the design of the tank bottom to supporting foundation does not justify the friction value above (e.g., leak detection membrane beneath the bottom with a lower friction factor, smooth bottoms, etc.). Alternatively, the friction factor is permitted to be determined by testing in accordance with Section 11.1.4.

- b. No additional lateral anchorage is required for steel tanks designed in accordance with reference documents.

- c. The lateral shear transfer behavior for special tank configurations (e.g., shovel bottoms, highly crowned tank bottoms, or tanks on grillage) can be unique and are beyond the scope of this standard.

15.7.6.1.6 Local Shear Transfer. Local transfer of the shear from the roof to the wall and the wall of the tank into the base shall be considered. For cylindrical tanks and vessels, the peak local tangential shear per unit length shall be calculated by

$$v_{\max} = \frac{2V}{\pi D} \quad (15.7-15)$$

- a. Tangential shear in flat-bottom steel tanks shall be transferred through the welded connection to the steel bottom. This transfer mechanism is deemed acceptable for steel tanks designed in accordance with the reference documents where $S_{DS} < 1.0g$.
- b. For concrete tanks with a sliding base where the lateral shear is resisted by friction between the tank wall and the base, the friction coefficient value used for design shall not exceed $\tan 30^\circ$.
- c. Fixed-base or hinged-base concrete tanks transfer the horizontal seismic base shear shared by membrane (tangential) shear and radial shear into the foundation. For anchored flexible-base concrete tanks, the majority of the base shear is resisted by membrane (tangential) shear through the anchoring system with only insignificant vertical bending in the wall. The connection between the wall and the floor shall be designed to resist the maximum tangential shear.

15.7.6.1.7 Pressure Stability. For steel tanks, the internal pressure from the stored product stiffens thin cylindrical shell structural elements subjected to membrane compression forces. This stiffening effect is permitted to be considered in resisting seismically induced compressive forces if they are permitted by the reference document or the Authority Having Jurisdiction.

15.7.6.1.8 Shell Support. Steel tanks resting on concrete ringwalls or slabs shall have a uniformly supported annulus under the shell. Uniform support shall be provided by one of the following methods:

- a. Shimming and grouting the annulus,
- b. Using fiberboard or other suitable padding,
- c. Using butt-welded bottom or annular plates resting directly on the foundation, and
- d. Using closely spaced shims (without structural grout), provided that the localized bearing loads are considered in the tank wall and foundation to prevent local crippling and spalling.

Mechanically anchored tanks shall be shimmed and grouted. Local buckling of the steel shell for the peak compressive force caused by operating loads and seismic overturning shall be considered.

15.7.6.1.9 Repair, Alteration, or Reconstruction. Repairs, modifications, or reconstruction (i.e., cut down and re-erect) of a tank or vessel shall conform to industry standard practice and this standard. For welded steel tanks storing liquids, see API 653 and the applicable reference document listed in Chapter 23. Tanks that are relocated shall be reevaluated for the seismic loads for the new site and the requirements of new construction in accordance with the appropriate reference document and this standard.

Table 15.7-4 Maximum Material Strength

Material	Minimum Ratio F_u/F_y	Max. Material Strength Vessel Material (%)	Max. Material Strength Threaded Material (%) ^a
Ductile (e.g., steel, aluminum, copper)	1.33 ^b	90 ^c	70 ^c
Semiductile	1.2 ^d	70 ^c	50 ^c
Nonductile (e.g., cast iron, ceramics, fiberglass)	NA	25 ^e	20 ^e

^aThreaded connection to vessel or support system.

^bMinimum 20% elongation per the ASTM material specification.

^cBased on material minimum specified yield strength.

^dMinimum 15% elongation per the ASTM material specification.

^eBased on material minimum specified tensile strength.

15.7.7 Water Storage and Water Treatment Tanks and Vessels

15.7.7.1 Welded Steel. Welded steel water storage tanks and vessels shall be designed in accordance with the seismic requirements of AWWA D100 with the following exceptions:

- a. The seismic design ground motion values shall be determined according to Section 11.4.b. Modify AWWA D100, Section 13.5.4.4 as follows:

13.5.4.4 Freeboard. Sloshing shall be considered in determining the freeboard above the MOL. Freeboard is defined as the distance from the MOL to the lowest level of roof framing. The freeboard provided shall meet the requirements of Table 29, ~~unless otherwise specified~~. The sloshing wave height ...

15.7.7.2 Bolted Steel. Bolted steel water storage structures shall be designed in accordance with the seismic requirements of AWWA D103, with the following exceptions.

- a. The seismic design ground motion values shall be determined according to Section 11.4.
- b. For Type 6 tanks, the overturning ratio, J , as determined using AWWA D103, Eq. (14-32) shall not exceed 0.785.

15.7.7.3 Reinforced and Prestressed Concrete. Reinforced and prestressed concrete tanks shall be designed in accordance with the seismic requirements of AWWA D110, AWWA D115, or ACI 350.3, except that the Importance Factor, I_e , shall be determined according to Section 15.4.1.1; the response modification coefficient, R , shall be taken from Table 15.4.2; the seismic design ground motion values shall be determined according to Section 11.4; and the design input forces for strength design procedures shall be determined using the procedures of ACI 350.3, except that S_{ac} shall be substituted for C_c in ACI 350.3, Section 9.4.2, using Eqs. (15.7-10) for $T_c \leq T_L$ and (15.7-11) for $T_c > T_L$ from Section 15.7.6.1.

15.7.8 Petrochemical and Industrial Tanks and Vessels Storing Liquids

15.7.8.1 Welded Steel. Welded steel flat-bottom, ground-supported petrochemical tanks, industrial tanks, and vessels storing liquids under an internal pressure less than or equal to 2.5 psig (17.2 kPa g) shall be designed in accordance with the seismic requirements of API 650. Welded steel flat-bottom, ground-supported petrochemical tanks, industrial tanks, and vessels storing liquids under an internal pressure greater than 2.5 psig (17.2 kPa g) and less than or equal to 15 psig (104.4 kPa g) shall be designed in accordance with the seismic requirements of API 620.

15.7.8.2 Bolted Steel. Bolted steel tanks are used for storage of production liquids. API 12B covers the material, design, and erection requirements for vertical, cylindrical, and above ground bolted tanks in nominal capacities of 100 to 10,000 barrels for production service. Unless required by the Authority Having Jurisdiction, these temporary structures need not be designed for seismic loads. If design for seismic load is required, the loads are permitted to be adjusted for the temporary nature of the anticipated service life.

15.7.8.3 Reinforced and Prestressed Concrete. Reinforced concrete tanks for the storage of petrochemical and industrial liquids shall be designed in accordance with the force requirements of Section 15.7.7.3.

15.7.9 Ground-Supported Storage Tanks for Granular Materials

15.7.9.1 General. The intergranular behavior of the material shall be considered in determining effective mass and load paths, including the following behaviors:

- a. Increased lateral pressure (and the resulting hoop stress) caused by loss of the intergranular friction of the material during seismic shaking;
- b. Increased hoop stresses generated from temperature changes in the shell after the material has been compacted; and
- c. Intergranular friction, which can transfer seismic shear directly to the foundation.

15.7.9.2 Lateral Force Determination. The lateral forces for tanks and vessels storing granular materials at grade shall be determined by the requirements and accelerations for short-period structures (i.e., S_{DS}).

15.7.9.3 Force Distribution to Shell and Foundation

15.7.9.3.1 Increased Lateral Pressure. The increase in lateral pressure on the tank wall shall be added to the static design lateral pressure but shall not be used in the determination of pressure stability effects on the axial buckling strength of the tank shell.

15.7.9.3.2 Effective Mass. A portion of a stored granular mass acts with the shell (the effective mass). The effective mass is related to the physical characteristics of the product, the height-to-diameter (H/D) ratio of the tank, and the intensity of the seismic event. The effective mass shall be used to determine the shear and overturning loads resisted by the tank.

15.7.9.3.3 Effective Density. The effective density factor (that part of the total stored mass of product that is accelerated by the seismic event) shall be determined in accordance with ACI 313.

15.7.9.3.4 Lateral Sliding. For granular storage tanks that have a steel bottom and are supported such that friction at the bottom to foundation interface can resist lateral shear loads, no additional anchorage to prevent sliding is required. For tanks without steel bottoms (i.e., the material rests directly on the foundation), shear anchorage shall be provided to prevent sliding.

15.7.9.3.5 Combined Anchorage Systems. If separate anchorage systems are used to prevent overturning and sliding, the relative stiffness of the systems shall be considered in determining the load distribution.

15.7.9.4 Welded Steel Structures. Welded steel granular storage structures shall be designed in accordance with the seismic requirements of this standard. Component allowable stresses and materials shall be per AWWA D100, except that the allowable circumferential membrane stresses and material requirements in API 650 shall apply.

15.7.9.5 Bolted Steel Structures. Bolted steel granular storage structures shall be designed in accordance with the seismic requirements of this section. Component allowable stresses and materials shall be per AWWA D103.

15.7.9.6 Reinforced Concrete Structures. Reinforced concrete structures for the storage of granular materials shall be designed in accordance with the seismic force requirements of this standard and the requirements of ACI 313.

15.7.9.7 Prestressed Concrete Structures. Prestressed concrete structures for the storage of granular materials shall

be designed in accordance with the seismic force requirements of this standard and the requirements of ACI 313.

15.7.10 Elevated Tanks and Vessels for Liquids and Granular Materials

15.7.10.1 General. This section applies to tanks, vessels, bins, and hoppers that are elevated above grade where the supporting tower is an integral part of the structure. Tanks and vessels that are supported by another structure are considered mechanical equipment and shall be designed in accordance with Section 15.3.

Elevated tanks shall be designed for the force and displacement requirements of the applicable reference document or Section 15.4.

15.7.10.2 Effective Mass. The design of the supporting tower or pedestal, anchorage, and foundation for seismic overturning shall assume that the material stored is a rigid mass acting at the volumetric center of gravity. The effects of fluid–structure interaction are permitted to be considered in determining the forces, effective period, and mass centroids of the system if the following requirements are met:

- The sloshing period, T_c , is greater than $3T$ where T = natural period of the tank with confined liquid (rigid mass) and supporting structure, and
- The sloshing mechanism (i.e., the percentage of convective mass and centroid) is determined for the specific configuration of the container by detailed fluid–structure interaction analysis or testing.

Soil–structure interaction is permitted to be included in determining T , provided that the requirements of Chapter 19 are met.

15.7.10.3 P-Delta Effects. The lateral drift of the elevated tank shall be considered as follows:

- The design drift, as determined by an elastic analysis, shall be increased by the factor C_d/I_e for evaluating the additional load in the support structure.
- The base of the tank shall be assumed to be fixed rotationally and laterally.
- Deflections caused by bending, axial tension, or compression shall be considered. For pedestal tanks with a height-to-diameter ratio less than 5, shear deformations of the pedestal shall be considered.
- The dead load effects of roof-mounted equipment or platforms shall be included in the analysis.
- If constructed within the plumbness tolerances specified by the reference document, initial tilt need not be considered in the P-delta analysis.

15.7.10.4 Transfer of Lateral Forces into Support Tower. For post-supported tanks and vessels that are cross-braced:

- The bracing shall be installed in such a manner as to provide uniform resistance to the lateral load (e.g., pretensioning or tuning to attain equal sag).
- The additional load in the brace caused by the eccentricity between the post-to-tank attachment and the line of action of the bracing shall be included.
- Eccentricity of compression strut line of action (elements that resist the tensile pull from the bracing rods in the seismic force-resisting systems) with their attachment points shall be considered.
- The connection of the post or leg with the foundation shall be designed to resist both the vertical and lateral resultant from the yield load in the bracing, assuming that the

direction of the lateral load is oriented to produce the maximum lateral shear at the post-to-foundation interface. Where multiple rods are connected to the same location, the anchorage shall be designed to resist the concurrent tensile loads in the braces.

15.7.10.5 Evaluation of Structures Sensitive to Buckling Failure. Shell structures that support substantial loads may exhibit a primary mode of failure from localized or general buckling of the support pedestal or skirt caused by seismic loads. Such structures may include single-pedestal water towers, skirt-supported process vessels, and similar single-member towers. Where the structural assessment concludes that buckling of the support is the governing primary mode of failure, structures specified in this standard to be designed to subsections a and b below and those that are assigned as Risk Category IV shall be designed to resist the seismic forces as follows:

- The seismic response coefficient for this evaluation shall be in accordance with Section 12.8.1.1 of this standard with I_e/R set equal to 1.0. Soil–structure and fluid–structure interactions are permitted to be used in determining the structural response. Vertical or orthogonal combinations need not be considered.
- The resistance of the structure shall be defined as the critical buckling resistance of the element, that is, a factor of safety set equal to 1.0.

15.7.10.6 Welded Steel Water Storage Structures. Welded steel elevated water storage structures shall be designed and detailed in accordance with the seismic requirements of AWWA D100 with the structural height limits imposed by Table 15.4-2.

15.7.10.7 Concrete Pedestal (Composite) Tanks. Concrete pedestal (composite) elevated water storage structures shall be designed in accordance with the requirements of AWWA D107, except that the seismic design ground motion values shall be determined according to Section 11.4.

15.7.11 Boilers and Pressure Vessels

15.7.11.1 General. Attachments to the pressure boundary, supports, and seismic force-resisting anchorage systems for boilers and pressure vessels shall be designed to meet the force and displacement requirements of Section 15.3 or 15.4 and the additional requirements of this section. Boilers and pressure vessels categorized as Risk Categories III or IV shall be designed to meet the force and displacement requirements of Section 15.3 or 15.4.

15.7.11.2 ASME Boilers and Pressure Vessels. Boilers or pressure vessels designed and constructed in accordance with ASME BPVC shall be deemed to meet the requirements of this section provided that the force and displacement requirements of Section 15.3 or 15.4 are used with appropriate scaling of the force and displacement requirements to the working stress design basis.

15.7.11.3 Attachments of Internal Equipment and Refractory. Attachments to the pressure boundary for internal and external ancillary components (refractory, cyclones, trays, etc.) shall be designed to resist the seismic forces specified in this standard to safeguard against rupture of the pressure boundary. Alternatively, the element attached is permitted to be designed to fail before damaging the pressure boundary provided that the consequences of the failure do not place the pressure boundary in jeopardy. For boilers or vessels containing liquids,

the effect of sloshing on the internal equipment shall be considered if the equipment can damage the integrity of the pressure boundary.

15.7.11.4 Coupling of Vessel and Support Structure. Where the mass of the operating vessel or vessels supported is greater than 25% of the total mass of the combined structure, the structure and vessel designs shall consider the effects of dynamic coupling between each other. Coupling with adjacent, connected structures such as multiple towers shall be considered if the structures are interconnected with elements that transfer loads from one structure to the other.

15.7.11.5 Effective Mass. Fluid–structure interaction (sloshing) shall be considered in determining the effective mass of the stored material, provided that sufficient liquid surface exists for sloshing to occur and that the T_c is greater than $3T$. Changes to or variations in material density with pressure and temperature shall be considered.

15.7.11.6 Other Boilers and Pressure Vessels. Boilers and pressure vessels designated Risk Category IV, but not designed and constructed in accordance with the requirements of ASME BPVC, shall meet the following requirements.

The seismic loads in combination with other service loads and appropriate environmental effects shall not exceed the material strength shown in Table 15.7-4.

Consideration shall be made to mitigate seismic impact loads for boiler or vessel elements constructed of nonductile materials or vessels operated in such a way that material ductility is reduced (e.g., low-temperature applications).

15.7.11.7 Supports and Attachments for Boilers and Pressure Vessels. Attachments to the pressure boundary and support for boilers and pressure vessels shall meet the following requirements:

- Attachments and supports transferring seismic loads shall be constructed of ductile materials suitable for the intended application and environmental conditions.
- Anchorage shall be in accordance with Section 15.4.9, whereby the anchor embedment into the concrete is designed to develop the steel strength of the anchor in tension. The steel strength of the anchor in tension shall be determined in accordance with ACI 318, Eq. (17.4.1.2). The anchor shall have a minimum gauge length of eight diameters. The load combinations including overstrength of Section 12.4.3 are not to be used to size the anchor bolts for tanks and horizontal and vertical vessels.
- Seismic supports and attachments to structures shall be designed and constructed so that the support or attachment remains ductile throughout the range of reversing seismic lateral loads and displacements.
- Vessel attachments shall consider the potential effect on the vessel and the support for uneven vertical reactions based on variations in relative stiffness of the support members, dissimilar details, nonuniform shimming, or irregular supports. Uneven distribution of lateral forces shall consider the relative distribution of the resisting elements, the behavior of the connection details, and vessel shear distribution.

The requirements of Sections 15.4 and 15.7.10.5 shall also be applicable to this section.

15.7.12 Liquid and Gas Spheres

15.7.12.1 General. Attachments to the pressure or liquid boundary, supports, and seismic force-resisting anchorage

systems for liquid and gas spheres shall be designed to meet the force and displacement requirements of Section 15.3 or 15.4 and the additional requirements of this section. Spheres categorized as Risk Category III or IV shall themselves be designed to meet the force and displacement requirements of Section 15.3 or 15.4.

15.7.12.2 ASME Spheres. Spheres designed and constructed in accordance with Section VIII of ASME BPVC shall be deemed to meet the requirements of this section, provided that the force and displacement requirements of Section 15.3 or 15.4 are used with appropriate scaling of the force and displacement requirements to the working stress design basis.

15.7.12.3 Attachments of Internal Equipment and Refractory. Attachments to the pressure or liquid boundary for internal and external ancillary components (refractory, cyclones, trays, etc.) shall be designed to resist the seismic forces specified in this standard to safeguard against rupture of the pressure boundary. Alternatively, the element attached to the sphere could be designed to fail before damaging the pressure or liquid boundary, provided that the consequences of the failure do not place the pressure boundary in jeopardy. For spheres containing liquids, the effect of sloshing on the internal equipment shall be considered if the equipment can damage the pressure boundary.

15.7.12.4 Effective Mass. Fluid–structure interaction (sloshing) shall be considered in determining the effective mass of the stored material, provided that sufficient liquid surface exists for sloshing to occur and that the T_c is greater than $3T$. Changes to or variations in fluid density shall be considered.

15.7.12.5 Post- and Rod-Supported Spheres. For post-supported spheres that are cross-braced:

- The requirements of Section 15.7.10.4 shall also be applicable to this section.
- The stiffening effect (reduction in lateral drift) from pretensioning of the bracing shall be considered in determining the natural period.
- The slenderness and local buckling of the posts shall be considered.
- Local buckling of the sphere shell at the post attachment shall be considered.
- For spheres storing liquids, bracing connections shall be designed and constructed to develop the minimum published yield strength of the brace. For spheres storing gas vapors only, bracing connection shall be designed for Ω_0 times the maximum design load in the brace. Lateral bracing connections directly attached to the pressure or liquid boundary are prohibited.

15.7.12.6 Skirt-Supported Spheres. For skirt-supported spheres, the following requirements shall apply:

- The requirements of Section 15.7.10.5 shall also apply.
- The local buckling of the skirt under compressive membrane forces caused by axial load and bending moments shall be considered.
- Penetration of the skirt support (manholes, piping, etc.) shall be designed and constructed to maintain the strength of the skirt without penetrations.

15.7.13 Refrigerated Gas Liquid Storage Tanks and Vessels

15.7.13.1 General. Tanks and facilities for the storage of liquefied hydrocarbons and refrigerated liquids shall meet the

requirements of this standard. Low-pressure welded steel flat-bottom, ground-supported storage tanks for liquefied hydrocarbon gas (e.g., liquefied petroleum gas or butane) and refrigerated liquids (e.g., ammonia) shall be designed in accordance with the requirements of Section 15.7.8 and API 620.

15.7.14 Horizontal, Saddle-Supported Vessels for Liquid or Vapor Storage

15.7.14.1 General. Horizontal vessels supported on saddles (sometimes referred to as “blimps”) shall be designed to meet the force and displacement requirements of Section 15.3 or 15.4.

15.7.14.2 Effective Mass. Changes to or variations in material density shall be considered. The design of the supports, saddles, anchorage, and foundation for seismic overturning shall assume that the material stored is a rigid mass acting at the volumetric center of gravity.

15.7.14.3 Vessel Design. Unless a more rigorous analysis is performed,

- a. Horizontal vessels with a length-to-diameter ratio of 6 or more are permitted to be assumed to be a simply supported

beam spanning between the saddles for determining the natural period of vibration and global bending moment.

- b. For horizontal vessels with a length-to-diameter ratio of less than 6, the effects of “deep beam shear” shall be considered where determining the fundamental period and stress distribution.
- c. Local bending and buckling of the vessel shell at the saddle supports caused by seismic load shall be considered. The stabilizing effects of internal pressure shall not be considered to increase the buckling resistance of the vessel shell.
- d. If the vessel is a combination of liquid and gas storage, the vessel and supports shall be designed both with and without gas pressure acting (assume that piping has ruptured and pressure does not exist).

15.8 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

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